Understanding the Eight-Point Algorithm and Rank-2 Enforcement

1. Why the Eight-Point Algorithm Uses 8 Points

- We want to find the fundamental matrix F that relates corresponding points between two views: $x^T{}_2F\,x_1 = 0$
  where $x_1 = [u_1, v_1, 1]^T$ and $x_2 = [u_2, v_2, 1]^T$ are homogeneous image coordinates in the two images.
- F is a 3×3 matrix with 9 entries, but it is only defined up to scale because multiplying F by any nonzero scalar λ does not change the epipolar constraint. Therefore, F has 8 independent degrees of freedom.
- Each pair of corresponding points provides one linear equation in the 9 unknowns of F. Thus, to solve for F (up to scale), we need at least 8 correspondences to get 8 independent equations.
- This ensures that the nullspace of the equation system $A f = 0$ is one-dimensional (unique up to scale).
- After obtaining F, we must enforce the geometric property that the true fundamental matrix has rank 2.
  This step is explained next.

2. Enforcing Rank 2

The fundamental matrix F must have rank 2 because it maps points to epipolar lines, and both the left and right nullspaces correspond to the epipoles in the two images. When we compute F numerically, noise can make it full rank (rank 3), so we must enforce rank 2.
To enforce rank 2:
1. Compute the Singular Value Decomposition (SVD) of the raw matrix F_raw:
  $F\_raw = U \Sigma V^T$
2. Set the smallest singular value $\sigma_3 = 0$:
  $\Sigma' = \mathrm{diag}(\sigma_1, \sigma_2, 0)$
3. Reconstruct: $F = U \Sigma' V^T$
This yields a valid fundamental matrix with rank 2.

Optionally, for numerical stability, normalized coordinates can be used before estimation. After solving, F is denormalized using the inverse of the normalization transforms for each image.

3. Why We Can Still Use 9 or More Points

- In theory, if we have 9 perfectly independent equations for 9 unknowns, the only solution to $A f = 0$ is the trivial one ($f = 0$).

- However, in practice, real correspondences are noisy, and the equations are only approximately valid. Instead of finding an exact nullspace solution, we solve a least-squares minimization problem: minimize $||A\ f||^2$ subject to $||f|| = 1$
  The solution is the right singular vector of A corresponding to its smallest singular value. This gives the best approximate solution for F even when there are 9 or more points.

- Thus, while 8 points are theoretically sufficient, modern implementations use many more points to obtain a robust estimate.
  Algorithms like RANSAC repeatedly sample minimal 8-point subsets to generate candidate F matrices and then refine them using
  all inliers.

4. Summary

Concept	Meaning
Number of entries in F	9
Defined up to scale	8 degrees of freedom
Each correspondence gives	1 linear equation
Minimum needed	8 points for exact solution
Rank 2 constraint	Enforced via SVD (set smallest singular value to 0)
>8 points	Overdetermined; solved via least-squares (SVD)
Practical use	8 points minimum, but typically 9+ with RANSAC or LSQ

Why Singular Value Decomposition (SVD) for finding Essential matrix?

SVD is the **most numerically stable and general-purpose** way to:

- Find the **nullspace** of a nearly singular or noisy matrix.
- Identify **dominant directions** and **discard small numerical noise**.
- Compute **low-rank approximations** (by setting small singular values to zero).

In particular:

Task	**Why SVD is used**
Solving $Af = 0$	SVD gives the vector in the direction of the smallest singular value → best approximate nullspace
Enforcing rank 2	SVD allows us to zero out the smallest singular value cleanly and reconstruct a lower-rank matrix
Handling noise	SVD naturally separates "signal" (large singular values) from "noise" (small ones)

Note: we can do similar steps for Essential matrix

Finding t and R from Essential matrix has been explained in lecture note 14.